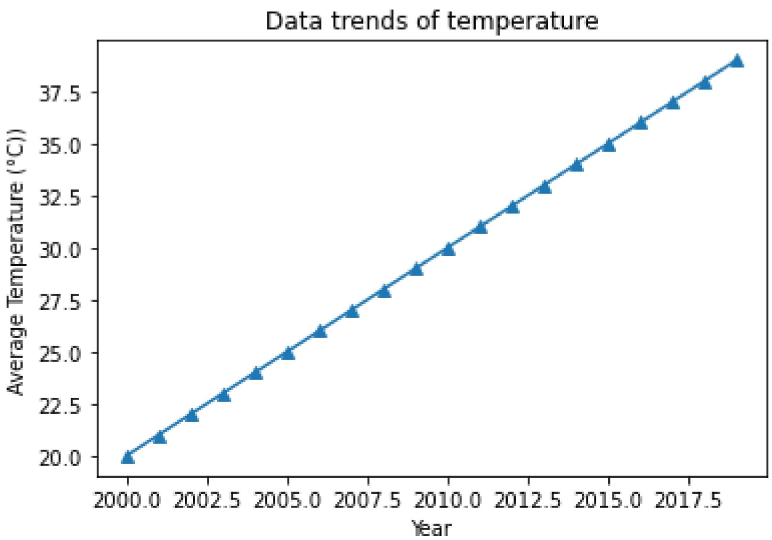
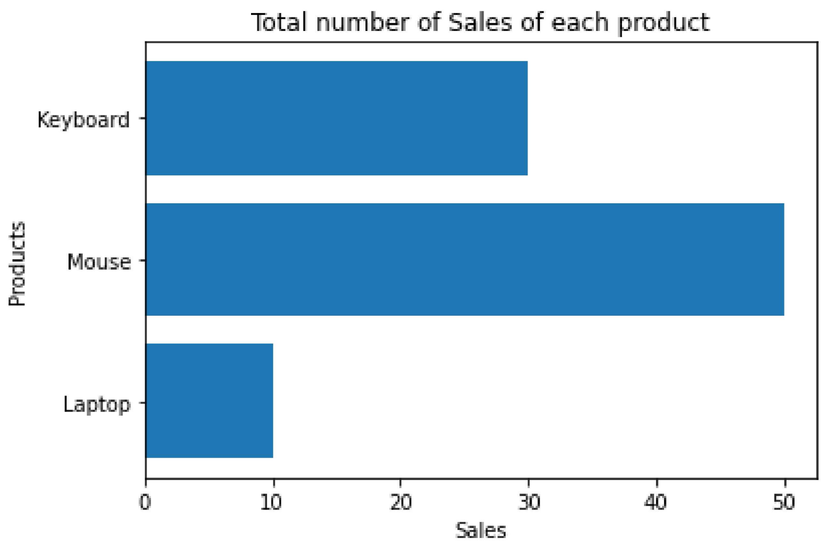
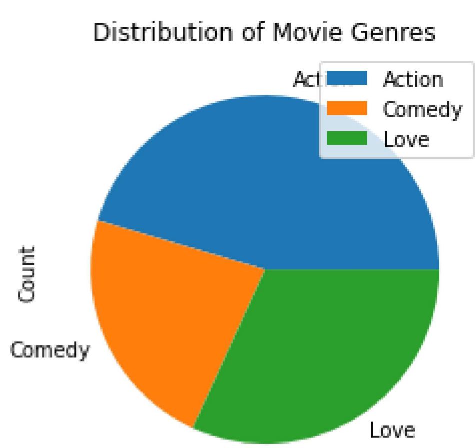
In [3]:
```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2000, 2020)),
'Temperature': list(range(20, 40))}
df = pd.DataFrame(data)
df.to_csv('weather.csv', index=False)
plt.plot(df['Year'], df['Temperature'], marker='^')
plt.title('Data trends of temperature')
plt.xlabel('Year')
plt.ylabel('Average Temperature (°C))')
plt.show()
```



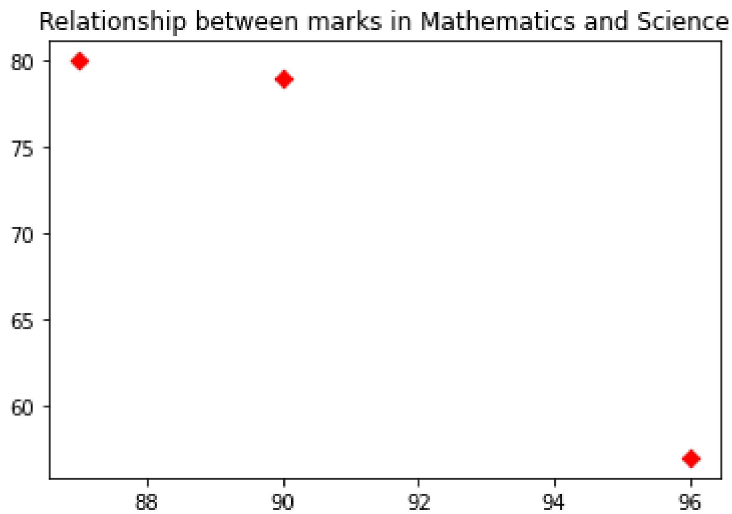
In [5]:
```
data1={
    'Products': ['Laptop', 'Mouse', 'Keyboard'],
    'Sales':[10, 50, 30]
}
df1=pd.DataFrame(data1)
df1.to_csv("sales.csv",index=False)
plt.barh(df1["Products"],df1["Sales"])
plt.title("Total number of Sales of each product")
plt.xlabel("Sales")
plt.ylabel("Products")
plt.show()
```



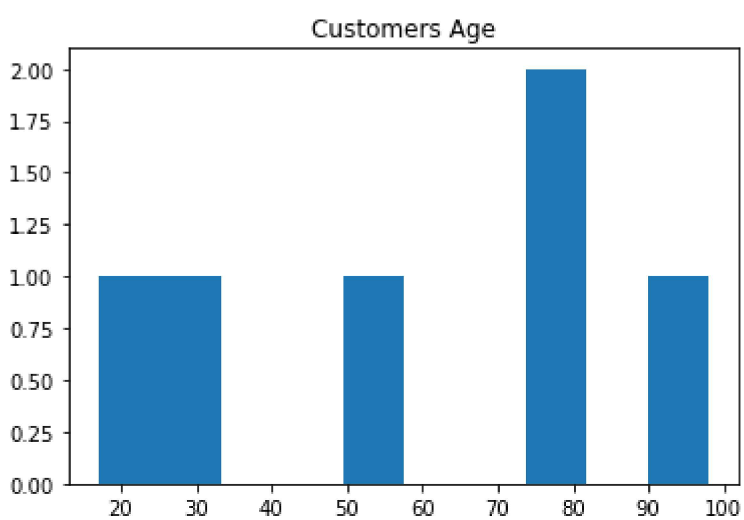
In [23]:
```
data2={
    'Genre': ['Action', 'Comedy', 'Love'],
    'Count':[10, 5, 7]
}
df2=pd.DataFrame(data2)
df2.to_csv("movies.csv",index=False)
df2.groupby('Genre').sum().plot(kind='pie', y='Count')
plt.title("Distribution of Movie Genres")
plt.show()
```



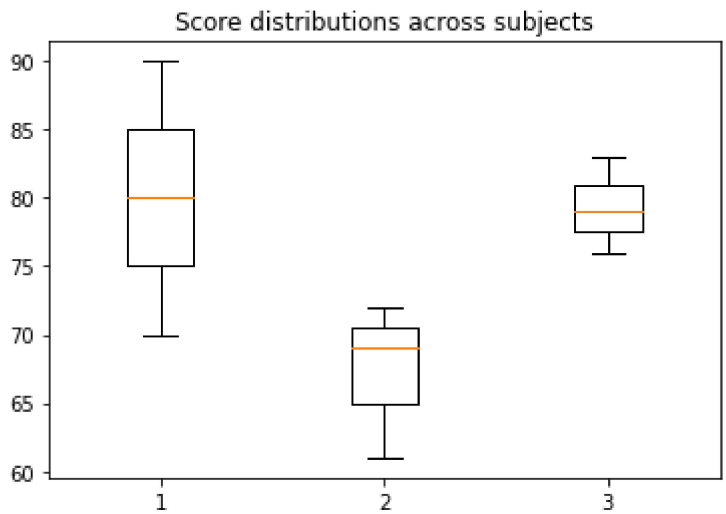
In [42]:
```
data3={
    'MathMarks': [87,90,96],
    'ScienceMarks':[80,79,57]
}
df3=pd.DataFrame(data3)
df3.to_csv("students.csv",index=False)
plt.scatter(df3['MathMarks'], df3['ScienceMarks'],color='red',marker='D')
plt.title("Relationship between marks in Mathematics and Science")
plt.show()
```



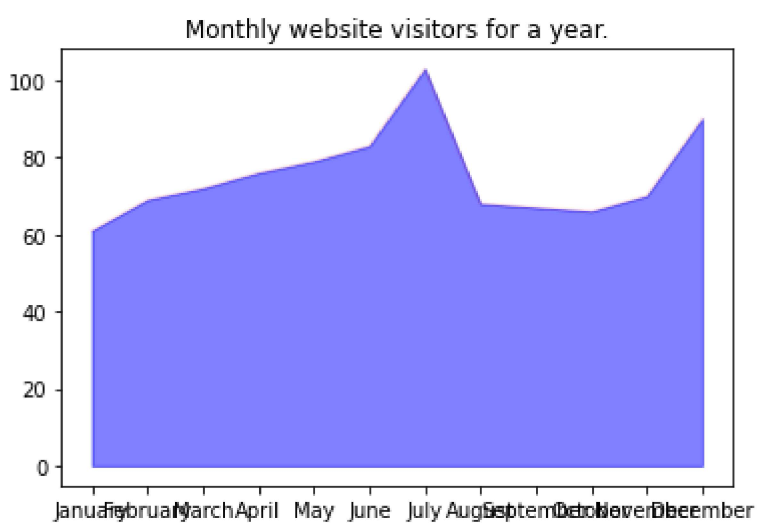
In [54]:
```
data4={
    'Age':[17,78,98,27,50,80]
}
df4=pd.DataFrame(data4)
df4.to_csv("customers.csv",index=False)
plt.hist(df4, bins=10)
plt.title("Customers Age")
plt.show()
```



In [55]:
```
data5={
    'Math':[70,80,90],
    'English':[61,69,72],
    'Science':[76,79,83]
}
df5=pd.DataFrame(data5)
df5.to_csv("scores.csv",index=False)
plt.boxplot(df5)
plt.title("Score distributions across subjects")
plt.show()
```



In [65]:
```
data6={
    'Month':['January','February', 'March', 'April', 'May', 'June', 'July', 'Augu
    'Visitors':[61,69,72,76,79,83,103,68,67,66,70,90]
}
df6=pd.DataFrame(data6)
df6.to_csv("Traffic.csv",index=False)
plt.fill_between(df6['Month'], df6['Visitors'], color='blue', alpha=0.5)
#plt.plot(df6['Month'], df6['Visitors'], color='red', alpha=0.1)
plt.title("Monthly website visitors for a year.")
plt.show()
```



In [70]:
```
data7={
    'Math':[78,85,89,74,92],
    'English':[88,79,94,90,96],
    'Science':[92,91,86,88,99],
    'History':[65,70,72,60,75],
    'Computer':[80,83,85,78,88]
}
df7=pd.DataFrame(data7)
df7.to_csv("students_scores.csv",index=False)
sns.heatmap(df7, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap of Student Scores')
plt.show()
```